# Ryan Baker

**Website** rybkr.com
**Email** hello@rybkr.com
**GitHub** github.com/rybkr
**YouTube** youtube.com/@rybkr
**LinkedIn** linkedin.com/in/rybkr

## Experience

**Database Software Engineer Intern** May 2026 – Aug 2026
*Oracle* *Redwood City, CA*

- Designed a resource-leak checker that identified several dozen processes failing to release resources during shutdown.
- Added an API option to disable report-generation timeouts, resolving missing report sections for the Exadata team.
- Refined per-table timeout policy when flushing performance data during Automatic Workload Repository (AWR) snapshots.

**Embedded Software Engineer Intern** May 2025 – Aug 2025
*Matrix Design Group* *Newburgh, IN*

- Shipped a deferred logging system for FreeRTOS that packed variadic arguments into compact byte buffers, offloading log processing from safety-critical calling threads.
- Parallelized CI jobs, cutting pipeline runtime by approximately 75%, from ~2 minutes to ~30 seconds per push.
- Built a Python tool using Tree-sitter to visualize finite-state machines, uncovering multiple bugs in production code.

## Projects

**Trillim AI** | trillim.com Dec 2025 – Present

- Co-developed a SIMD-optimized inference engine, achieving ~1.5–2× the speed of Microsoft's BitNet in benchmark testing.
- Enabled local inference for both language and image-generation models, filling a gap in llama.cpp's model support.
- Built a Python SDK and WebAssembly integration to enable model inference in Python applications and web browsers.

**GitVista** | gitvista.io Oct 2025 – Present

- Built a Git engine in Go and a repo visualizer for browsing history, inspecting files at any revision, and analyzing repo activity.
- Prioritized backend-to-browser data delivery around the active view to support exploration of large repositories.

**Totally** | totally.sh Jun 2026 – Present

- Built a Go CLI that estimates OpenAI and Anthropic agent costs from local transcripts using updated model pricing.
- Implemented cost breakdowns by model, session, time period, and provider to help developers track usage.

## Education

**M.S.E. Electrical & Computer Engineering**, Purdue University Aug 2026 – May 2027
*Concentration in Artificial Intelligence* *West Lafayette, IN*

**B.S.E. Computer Engineering**, Purdue University Aug 2023 – May 2026
*Concentration in Computer Systems* | GPA: 3.5 of 4.0 *West Lafayette, IN*

## Teaching & Speaking

**Technical Educator** | youtube.com/@rybkr Feb 2024 – Present

- Create educational content on Unix, low-level programming, and AI systems for 28K+ subscribers, with 1M+ views.

**Teaching Assistant** | Computer Networks Aug 2026 – Present
*Purdue University* *West Lafayette, IN*

- Support 60 students in networking concepts with weekly office hours covering internet architecture, routing, and TCP/IP.
- Help students debug C and Python assignments involving socket programming, HTTP, and distributed protocols.

**Course Instructor** | Intro to Modern C++ Jan 2025 – May 2025
*Purdue IEEE* *West Lafayette, IN*

- Developed and taught a Modern C++ course for 40 students through Purdue IEEE, certifying successful students.
- Delivered weekly 90-minute lectures from C++ fundamentals through templates, compile-time programming, and safety.

**Conference Speaker** | Core C++ Group Nov 2024

- Presented "Messing with Floating Point": a talk on floating-point use in safety-critical and performance-critical systems.

**Teaching Assistant** | Linux & Bash Aug 2024 – Dec 2024
*Purdue IEEE* *West Lafayette, IN*

- Assisted with course content development; helped students with assignments on Linux fundamentals and Bash scripting.